Impact and Fall Detection from Instrumented Rugby Vest and Mouthguard

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Declaration

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Abstract

This research aimed to utilise multiple wearable sensors, which include the instrumented rugby vest and instrumented mouthguard, to analyse rugby team performance and player safety and well-being, as well as the usability of the instrumented rugby vest prototype. Six CNN models were developed and individually trained as learners. Three of each were used to detect impact events, while an additional three were used to detect falls. All exceeded 95% accuracy in both validation and testing. The ensemble learning technique was applied to these learners in an attempt to improve their accuracy, however, the final accuracies did not meet expectations due to the limited and less diverse training dataset. All outputs were presented as a final result in the form of a dashboard visualisation containing data and an interactive graph. This result achieves the objective of this study as the rugby coach can use it to monitor player impact and make prompt player substitutions. Numerous topics pertaining to future work were discussed, including the findings for the next design of the instrumented rugby vest to enhance it along the way to an actual product.

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Acknowledgements

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Chapter 1

Introduction

In the current age of big data, there is extensive utilisation of modern electronic devices such as smartphones and wearable gadgets for the purpose of gathering and generating substantial volumes of technical data. Numerous industries leverage the data obtained by these devices for diverse objectives, including the analysis of consumer behaviours within the marketing domain [15], the enhancement of medical evaluation and treatment procedures within the healthcare field [10, 43], or the advancement of teaching and learning methodologies within the education sector [33]. In addition to this, they are also utilised extensively in the realm of athletics to gather data from athletes during training sessions and competitive matches for purposes, such as monitoring exercise routines [1, 8, 36], evaluating athletes' performance [7, 31, 47, 52], and quantifying the intensity of impact that may lead to injuries, particularly in contact sports such as rugby [11, 22, 23, 24, 25, 26, 27, 28, 29, 40, 49, 51, 53, 54, 55]. It is not similar to other sports in common as rugby is a sport with a significant risk of injury due to its reliance on a substantial level of core strength muscles, e.g., lower back muscles or transverse abdominis [4], as well as a combination of skills, e.g., running, passing and tackling [30], in order to defeat the opponent while keeping the players safe.

Rugby is a widely recognised and popular contact sport, with a substantial global player base of over 8 million individuals affiliated with the World Rugby international federation [57]. Given that it is a contact sport, the primary factors contributing to injuries are the impact and falls resulting from the various motions involved. Tackling is widely acknowledged as a prominent factor in causing injuries due to its significant impact on both the one executing the tackle and the individual being tackled, prompting the Rugby Football Union to introduce a new regulation aimed at enhancing player safety via the use of lower tackle heights [46]. Regarding a study that involved the analysis of video recordings from 434 professional rugby matches, it was observed that the average number of tackle events per match was 203 ($\pm$ 29). These tackle events resulted in a total of 1,559 injury assessments, which can be further categorised into 1,348 injuries that necessitated on-field treatment and 211 injuries that required player replacement [44]. Yet, another study revealed that there are factors causing impacts with higher risk of injury per event to the player than tackles, which are collisions and scrums, with 70% and 60% chances higher than tackles, respectively [18]. At the same time, many injuries are the result of falls. A study was conducted to provide an understanding of the relationship between falls and injury, and it revealed that various types of injuries that can occur as a result of falls, such as Sternoclavicular and Glenohumeral joint injuries. These injuries typically occur when players fall directly onto their shoulder. Additionally, Brachial plexus traction injuries were identified when individuals fall forward and downwards, resulting in simultaneous impact of the head and shoulder with the ground.

Researchers have continued to initiate analysis to improve safety in players from time to time by incorporating technology into their studies, such as using wearable motion sensors for analysing professional players' movement, which can be used to assist coaches in designing training programmes in the future [11], using an instrumented mouthguard to analyse collision characteristics, such as intensity, in both linear and rotational acceleration, which can be used to determine a player's replacement in order to boost the intensity of the game [54] or even using force sensors to analyse tackle impact force to specialised shoulder tackle techniques [49]. Based on these results, coaches have the opportunity to use this information to optimise team plans, while simultaneously ensuring the safety of players during rugby games.

Nevertheless, the findings of these studies were derived from a singular device, so presenting a potential bias and limited perspectives to the researchers. In addition, several studies have been explicitly aimed at analysing the occurrence of impacts and collisions, since they are well recognised as significant causes of injury. However, it is important to take into account that falls may also result in severe injuries. Based on the identified needs and limitations, this research project will be carried out.

1.1 Project Aim

Refer to the Appendix A.1.

1.2 Project Objectives

Refer to the Appendix A.2.

1.3 Project Plan

Refer to the Appendix A.3.

1.4 Project Requirements

Refer to the Appendix A.4.

Chapter 2

Literature and Technology Survey

In brief, the main aims of this study are on improving team performance and reducing rugby player injuries, while simultaneously conducting a usability evaluation of the instrumented rugby vest prototype, which has been produced by the University of Bath. The first stage of this study started with the utilisation of wearable sensor technology to collect data on the movement and impact of rugby players. Then, the data is analysed with machine learning models to obtain insights that can be used to predict the occurrence of impact and fall events, which are the most likely causes of injuries in this sport. Finally, visualise the results for substantial benefits. Due to this project's principal goals, the literature review has been thoroughly performed under the following topics.

The following discussion begins with the use of wearable sensors in rugby, with the aim of providing context for which sensors have been utilised for achieving which goals and highlighting the wearable sensors utilisation gap in previous research. This is followed by a discussion regarding the benefits associated with the use of machine learning algorithms, along with an exploration of the appropriate model architecture identified in previous research that may be applied to this particular dataset and a potential application of an analytical reporting format that meets the purpose of this study accordingly. The study will conclude with a discussion of the machine learning database's definition, as well as its advantages and disadvantages as it could be integrated for greater future benefits. This discussion may reveal further benefits for other research using similar analytical methodologies.

2.1 The Application of Wearable Sensors in Rugby

In prior studies, different types of sensors have been utilised to collect data from athletes according to their purposes such as the motion sensors which are primarily used for monitoring movement, e.g., pedometers, accelerometers or GPS sensors or the physiological sensors which are used for monitoring the workload and fatigue, e.g., heat flux or heart rate sensors [31]. But, since head injury is a serious concern in rugby, the common theme that appeared in almost every rugby-related research is to monitor the head impact.

The instrumented mouthguard manufactured by various brands, as exemplified in Figure 2.1, have frequently been utilised as a wearable sensor in such studies owing to its dual functionality of providing protection and capturing head acceleration events of players, as well as monitoring head impacts. Those studies either conduct head kinematic analysis with a single instrumented mouthguard system [26, 40, 53, 54] or perform sensor accuracy comparisons among multiple instrumented mouthguard systems [23] or between a single instrumented mouthguard system with other head wearable sensors, e.g., helmet sensors [25]. Aside from that, other wearable sensors were also used to address a similar issue but with various forms of analysis, such as automatically detecting and counting tackles using the instrumented harness [22, 24] or detecting head impact location using a skin patch sensor [29].

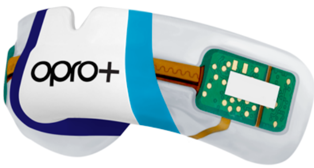

Figure 2.1: Example of the wearable sensors, which is a customised instrumented mouthguard (OPRO+), developed by OPRO International Limited [37]

Meanwhile, other sports studies offered a broader range of sensor usage, which could demonstrate more capabilities of these sensors than the rugby-related studies carried out. For example, the running and jumping performance was assessed using wearable motion sensors [52] or the daily activities were predicted using accelerometers and GPS sensors [16]. These outcomes are not exclusive to any sport and may also be applied to non-athletes as well.

These findings highlight a gap in rugby-related research that allows researchers to delve deeper by integrating several sensors with various objectives to obtain results in a broader perspective which aligns with this project's scope as it is a combination between the instrumented rugby vest and instrumented mouthguard. An example of a scope that might be undertaken in subsequent research is a real-time evaluation of rugby player performance and prediction of injury level in match games using the instrumented mouthguard and the motion sensors by determining the correlation between performance reduction and the number of collisions which may help to prevent significant injury in athletes and determine the proper treatment programme.

2.2 The Application of Machine Learning Algorithms

Due to the capability of wearable devices these days, they could generate massive data in a millisecond and transfer them to computing devices in real-time. However, not all data could be used. These data may contain good quality data that fit your purpose, good quality data that does not align with your project scope or distorted data. For this reason, the data processing steps by using the machine learning algorithm are required.

Most previous studies performed head impact event analysis without utilising the machine learning algorithm. They simply defined the threshold for linear acceleration at 10g [26, 27, 54]. This indicates that data was preserved only when the accelerometer exceeded the linear acceleration threshold, otherwise, the data were discarded. This threshold value has been widely defined at 10g as it has been studied to be a suitable level for differentiating between contacting and non-contacting events [28]. However, another study found that contacting events had an expansive range of linear acceleration. Up to 75.7%

of the head impact from contacting events was found less than 10g or approximately 5g. Therefore, the data was mingled with a non-contact event [53]. This finding demonstrates the limitation caused by a single data source, which requires a solid value to be set as a threshold for cleaning up. Although, as mentioned in the previous section, integrating with data from other sensor may overcome this issue since the researcher would have a broader perspective to identify the missing event owing to the threshold. Yet, adopting a machine learning model to categorise the data tends to be a long-lasting solution.

To solidify this assumption, there is an analysis found using the mouthguard to gather data for studying the head acceleration event classification algorithm. The CatBoost, eXtreme Gradient Boosting (XGBoost), Support Vector Machine (SVM), and Adaptive Boosting (Adaboost) classification algorithms were used in the study to detect impact events and the results were then compared, and as a result, the CatBoost achieved the most accuracy in the optimum phase, while the SVM classifier received the highest accuracy in the test phase [40]. Another study, on the other hand, was conducted to classify the motions of rugby players, including standing, walking, jogging, sprinting, and etc. The data was gathered through motion sensors in the form of time-series data, which was afterwards analysed adopting the Long short-term memory (LSTM) algorithm. As a consequence, the model exhibited a predictive accuracy of up to 80.40% [11]. Based on the findings of these studies, it is shown that the adoption of machine learning algorithms in the process may provide advantages by minimising the loss of valuable information.

2.3 The Convolutional Neural Network Architecture

During the literature review conducted on the previously discussed subject, there is a study suggested to explore the utilisation of Convolutional Neural Network (CNN) architecture in deep learning models for sport player motion detection [11]. This leads to a review of the relevant literature on this topic.

Among all techniques used in the creation of machine learning models, the deep learning technique is one of those approaches and the CNN stands out as the most popular architecture [3]. The architecture of the CNN may be categorised into three distinct types, which are determined by the directional movement of the kernel. The one-dimensional convolutional neural network (1D CNN) is mostly used for analysing time-series data, but the two-dimensional convolutional neural network (2D CNN) and three-dimensional convolutional neural network (3D CNN) have shown outstanding results in tasks related to image recognition. As shown in Figure 2.2, the CNN architecture is designed to consists of several layers, including the input layer, the convolutional layer, the pooling layer, and the fully-connected layer. The input layer holds the initial information of the input. The convolutional layer is responsible for extracting features by sliding a kernel or filter across the data. The output of this layer is influenced by three hyperparameters, which include the size of a filter, the stride, and the zero-padding. The pooling layer is used for downsampling the feature maps, which is capable of using either average pooling or max pooling methods. Average pooling is a method that calculates the average of elements within a region of the feature map covered by the filter, while max pooling method selects the maximum element. Finally, the fully-connected layer, also known as the dense layer, serves as both a hidden layer and an output layer. It encompasses weights, biases, and neurons to effectively generalise the extracted features [2, 38].

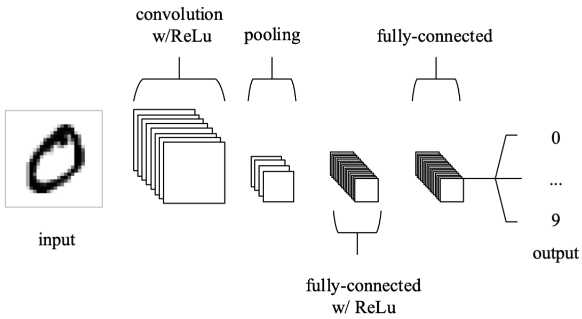

Figure 2.2: Basic CNN architecture [38]

Initially, the CNN was primarily designed for image processing. Subsequently, researchers conducted more explorations into the use of the CNN with other types of data, through which it shown remarkable performance, particularly with time series forecasting [5, 56]. Moreover, there is a research conducted a pooled analysis of 52 sport-specific movement recognition studies and discovered that although the SVM method was often used in the majority of these research, those studies that utilised the CNN approach shown outperformed the others [9]. Based on the findings presented, the CNN has considerable potential as a feasible choice for implementation in this research study.

2.4 The Utilisation of Dashboard

Reporting the results of the analysis is essential in order to assist those who matter to understand the context of findings and determine further actions. However, the majority of sport analysis studies discovered throughout the literature review conducted for this research only presented their findings in an academic manner, rather than in a user-friendly style that could be easily understood and used by people engaged in real-world situations. Hence, the researcher identified this as an interesting research gap to be addressed in the current study.

Data visualisation is a powerful tool for conveying complicated information in a visual form to audience [32]. To create the efficient data visualisation, its type and design should be carefully considered to answer the needs. One famous type of the data visualisation is known as a dashboard which is widely used in the business sector to facilitate the monitoring and analysis of multiple operations, allowing users to rapidly comprehend information such as product sales trends, team performance, and budget management. With a well-designed dashboard, users could save time creating and maintaining multiple reports, thus enhancing their total productivity and operating at higher levels of performance. [45]. Managing a sports team is similar to managing a business. In order to effectively operate and bring the win to team, it is necessary to possess a clear vision, recognise and mitigate weaknesses, identify opportunities for performance enhancement, and eventually achieve success. With this similarity, dashboards may also be effectively used in sport sector due to its ability to convey information in a fast and efficient way, as some sports are expected for all parties involved to exhibit promptness in their actions and decision-making processes in response to the fast pace of the game. Rugby is also classified under that particular group. As a result of this observation, the researcher finds it advantageous to further explore in this study.

In the context of design, it is essential to prioritise the functionality of the dashboard. If the design of a dashboard is insufficient, it is likely to be disregarded by users and judged ineffective. This is a common issue seen in many dashboards, where an excessive amount of information is included in an attempt to maximise content. Consequently, this approach often results in overwhelming users, making it difficult for them to navigate and comprehend the important details [17]. In order to optimise its effectiveness, it must be mindful to take into account the pre-attentive features, which are the attributes that trigger the pre-attentive cognitive processes of the brain. These attributes include length, orientation, and colour [32]. According to a study, however, designing it with a single attribute is sufficient, whereas adding multiple attributes does not aid in its improvement because too many attributes to process require the brain to perform a sequential processing. Whilst, colour was found to be an important attribute to have resulted in a significant effect on attracting attention and interpreting data [20], other attributes could be investigated further in future studies.

2.5 The Integration to Machine Learning Database

Machine Learning Database (MLDB) is initially designed and created by Datacratic in 2016 and sold to Element AI in 2017 [34]. It is an open-source system developed especially for machine learning applications [34], providing a centralised, high-speed, and scalable platform for big data machine learning operations such as storing data, cleaning up data, transforming data, training machine learning models, and deploying machine learning models. A schema-less SQL database is provided for users in order to upload and store data with flexibility and high efficiency in the form of a 3D sparse matrix containing named rows, named columns, timestamps, and values, as well as built-in machine learning features that may cope with basic to advanced algorithms such as deep learning or natural language processing [35]. The overall process flow is shown in Figure 2.3.

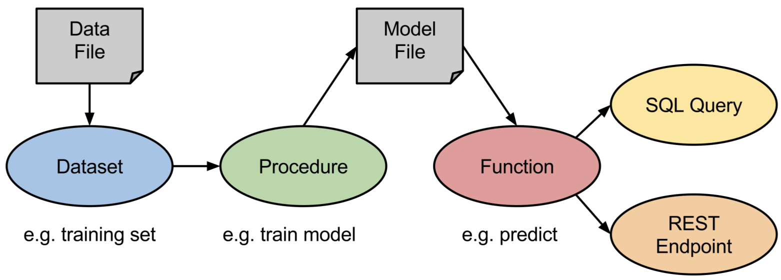

Figure 2.3: Machine learning database process flow, designed by Datacratic company [34]

However, there is a drawback for every advantage. According to the study, despite all of the benefits of this concept, it still has a disadvantage since it does not save user operations throughout the pre-processing cycle. This implies that the MLDB does not track user modifications to feature sets and hyperparameters [50]. Having this data could provide users insights in terms of comparing each modification, which may assist users in effectively enhancing the machine learning model performance. In the lack of this information, users have to put in significant effort and time in order to carry out the modelling, determine the best-fit parameters, or even apply trial-and-error experience to a new analysis.

Even though the company's product was not widely adopted, the concept of this product was replicated in some studies due to its design and performance, for example in studies where the objective was to utilise manufacturing chips in the research in which a massive amount of data must be generated in order for the machine learning model to predict data patterns [48, 59]. Consequently, the concept of the MLDB is considered a perfect match system for integrating with wearable sensors in this sports-related study, as each sensor gives a huge quantity of data in a short period of time. Therefore, it is interesting that only a few research papers adopting either this product or this concept had been found while performing a literature review. This finding contradicts its broad range of capabilities. And, with a relatively small amount of articles, there is a gap in expertise that may be filled with further studies.

Chapter 3

Methodology

To achieve the desired objectives of this study, two primary deliverables were necessary: the analysis results and the visualisations. These outputs aimed to improve team performance and minimise player injuries, while also evaluating the usability of the instrumented rugby vest. Following a thorough review of the existing literature, many research gaps were identified. In an effort to address these deficiencies, the present study was conducted, using the methods outlined in the subsequent discussion. This project was conducted in collaboration with the Department for Health, together with a skilled technician, of the University of Bath.

3.1 Data Collection

Due to the purposes of this study, data regarding the movement of players and the levels of impact pressure they experience will be gathered. The data collection approach used in this research was designed to maximise participant safety and optimise the study's advantages simultaneously. Therefore, the approach was designed to gather data by allowing a participant to wear wearable sensors and execute a tackle on a tackle bag. Data was designed to colllected in the form of a time-series dataset in order to get meaningful insights on the movement patterns of rugby players. This approach was used since it allows for the study of consecutive data points that were recorded over a certain time period. The selection of tackling as the primary focus of this research is justified by its widely recognised correlation with the risk of injury [44]. By thoroughly investigating this movement, the analysis could provide valuable insights that may contribute to the improvement of player safety and overall well-being during gameplay. Regarding the tackle bag, the decision to use it was based on its frequent incorporation in rugby training sessions [21]. In addition, tackling that involves direct physical contact between players may result in injuries, which could be avoided by using this. However, the punch bag was utilised as the tackle bag owing to equipment limitations in this study, while a participant safety was preserved. Moreover, it is necessary that the individual chosen for this research is a rugby player who fully understand how to execute tackles safely. Due to this, the data collection for this study was conducted by a rugby player who is a member of this project's team. Throughout the process, this data collection was conducted inside a controlled laboratory setting, ensuring strict compliance with safety protocols and the preservation of a member who participate confidentiality. No personally identifiable information was collected.

This study used two unique types of wearable sensors across three different body regions to enhance the coverage of data collection and mitigate the risk of missing data that may fall beyond the range of each sensor or fail to meet the predetermined threshold. This approach was adopted to address the gaps identified in the existing literature. The first wearable device under consideration is the instrumented mouthguard in Figure 3.1, developed by Prevent Biometrics company. The purpose of this mouthguard is to detect instances of head impact that above a threshold of 5g, particularly by measuring both linear and angular acceleration and velocity, and the workload of the collisions, as well as identifying the specific location of the head that is affected [41].

Figure 3.1: The instrumented mouthguard as the first wearable sensor, developed by Prevent Biometrics company [42]

The second item under consideration is the instrumented rugby vest shown in Figure 3.2, which was designed and made by a technician of the University of Bath. The instrumented rugby vest is a rugby shirt that was modified to include 32 force-sensing resistors (FSRs), in Figure 3.3, and the MPU-6050 motion tracking sensor, in Figure 3.4. Additionally, Global Positioning System (GPS) was integrated, however, it was not used in this study because data collection was conducted in an indoor place, which rendered GPS unable to function. All of these electronic components were connected via a microcontroller, allowing this wearable sensor to capture data and transmit it to a computer via wireless network. This wearable sensor allows the monitoring and reporting of pressure applied to FSRs in Newton units from two distinct regions [14], which are the shoulder and chest areas. Each of these places was equipped with 16 FSRs, arranged in a 4 by 4 configuration for the front shoulder area and a 2 by 8 configuration for the front chest area. This setup facilitates precise and reliable measuring capabilities. In addition, the MPU-6050 sensor that located at the rear waist level of this instrumented rugby vest was equipped with a tri-axial accelerometer and a tri-axial gyroscope which helps this wearable device to also capable of capturing and recording data pertaining to angular acceleration and velocity [13], similar to an instrumented mouthguard.

To begin the data collection process, the punch bag was strategically placed on a sliding-rack in order to simulate the actual scenario of an adversary player quickly approaching towards the individual engaging in the activity. Simultaneously, the team member proceeded to approach the punch bag in order to execute the tackle movement. This team member was intentionally arranged the starting point at different positions inside the laboratory to collect data from multiple angles. During tackling, the FSRs and other devices on the instrumented rugby vest, as well as the instrumented moutguard, will capture the force pressure and other impact-related data as previously described. The

Figure 3.2: The instrumented rugby vest (front and back) utilised as the second wearable sensor, developed by the University of Bath

Figure 3.3: Force-Sensing Resistors embedded in the instrumented rugby vest [14]

obtained data was promptly transmitted to the computer in real-time. Furthermore, the video footage was blurry recorded throughout the process of data collection. This will be used in a later stage for the purpose of labelling data in order to facilitate the comparison and validation of actions performed at different points in time. As a result, a total of 14 tackles were performed on a punch bag. The data from the instrumented mouthguard was retrieved in a readily usable format in a Comma Separated Values file (.csv), while the data from the instrumented rugby vest was in its raw form and was provided in the format of a Text file (.txt). The reason for this is because the data obtained from the instrumented mouthguard was automatically processed by the company's software in order to identify head impact events and filter out any noise or events that fail to exceed a specified threshold, requiring no further processing.

3.2 Data Preparation

Upon the successful acquisition of data, the preparatory stages are necessary in the purposes of data cleansing and transformation. This phase was designed to begin with the import of input files as datasets and continue until the inputs have been formatted in the appropriate shapes and types. The outcomes of this phase are processed to be applicable for input through machine learning algorithms with the aim of identifying occurrences of impact and falls, as described in this study. However, this stage and the succeeding

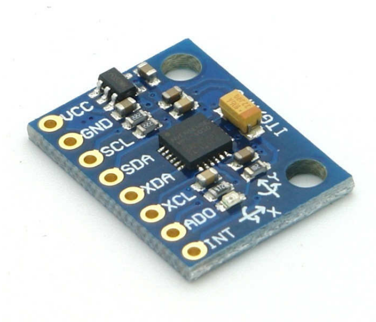

Figure 3.4: MPU-6050 Motion tracking device embedded in the instrumented rugby vest [13]

stage was designed only for the data obtained from the instrumented rugby vest. The data from the instrumented mouthguard was then combined in the final stage due to a reason stated in the data collection section.

The initial step of this stage involves the process of reading all input files and then transferring the data into structured tables. Individual data tables were created for each file, and combining them is not possible due to the disconnected time frames involved, which may result in incorrect information. Once data had been transferred, a hardware problem was observed. Due to the fact that this instrumented rugby vest was a prototype, the FSR positions and data column arrangement were not well-aligned which leads to confusion. To effectively manage the data in this layout, it is necessary to include the process of rearranging pressure data columns afterwards.

The data cleaning process was then implemented as a next step in order to resolve incorrect values. These inaccuracies arose from the unreliable FSRs, which were prone to short circuits resulting from the overlapping wires originating from either hand stitching or body movement. To mitigate the issue, pressure data points exceed certain thresholds were excluded. These pressure thresholds were defined at a minimum value of 100 and a maximum value of 1023, as informed by the technician engaged. To elaborate, the range of pressure values collected by the resistors is between 0 and 1023. Under a steady state, the resistors tend to settle within the range of 800 to 1023 and the resistance value decreases according to the amount of pressure applied. However, the range of 0 to 100 will be classified as circuit faults in this study due to the fact that the average impact level of tackles performed in this data collection is not a high level impact in order to mitigate any potential injuries that may occur to the team member. Moreover, additional thresholds were also informed to define for the acceleration and rotation data obtained from the accelerometer and gyroscope, which are $\pm$ 8 times of gravity and $\pm$ 500, respectively. The example pattern of the data after it was properly cleaned is shown in Figure 3.5, which displays the average values of FSRs in the shoulder area and chest area, as well as the data obtained from the accelerometer and gyroscope in the MPU-6050 motion tracking device. The positive and negative peaks in the figure correspond to occurrences of tackle event, including both falls and non-falls.

Due to the potential future application of the machine learning models developed in this study, which could be used with the improved design of the instrumented rugby vest that has already addressed the issue of short circuits, it is necessary to perform the data manipulation process by substitute those incorrect pressure values with filled-in values

Figure 3.5: Sample graph generated from instrumented rugby vest data (before). This displays the data points for which inaccuracies were set to 0 to exclude.

after rectifying the faults. Hence, the incorporation of these values may help the learning process of the models in preparing for the advanced stage of the instrumented rugby vest design. In order to calculate the appropriate filled-in values, the small size of the resistors and their close proximity to one another were considered as it is not feasible for just one resistor to be selectively targeted during rugby match, such tackling, scrumming, or being struck by the ball. Regarding this justification, these values are derived by calculating the mean of the resistors surrounding the target resistor.

Besides, it is required to have the expected outcomes alongside the input for the purpose of facilitating the learning process of machine learning models. Therefore, a manual assignment of labels was performed to identify the presence or absence of impact or fall movements in each data point or segment. These labels are also known to as ground-truth labels. In order to categorise the data and provide a label, the data pattern was analysed through the use of graphical representations as seen in Figure 3.5 and utilising the video recording as a guidance in order to extract the correlation between the observed data patterns and the movements performed by those who took part. The first step of the analysis included simplifying the pressure data by substituting each value with the mean value during a time interval of 0.25 seconds (6 records or rows). The first graph in Figure 3.6 illustrates how this aids in the apparent accessibility of data patterns. The vector magnitude of the accelerometer data was also determined, as shown in the second graph in Figure 3.6, by the use of a Mathematical Equation 3.1.

$$Magnitude = \sqrt{X^2 + Y^2 + Z^2} \tag{3.1}$$

Next, the threshold values for identifying both events were chosen while assessing the correlation between the two sources. The value of 400 was determined to represent the impact event in the pressure values, whilst the value of 25 represented the fall event in the magnitude values. The initial expected results for each row was defined according to predetermined thresholds. A value of 1 (flag 1) indicates the presence of an event, while a value of 0 (flag 0) indicates the absence of an event. These outcomes were sorted depending on the type of event, which are shoulder impact, chest impact, impact (a combination of shoulder and chest impacts), and fall events. The fault flag may occurred due to the presence of data fluctuations, which resulting in the occurrence of differential assigned flag at the millisecond level. This issue was rectified by reassigning the event flags between two identified events of the same type, if their time difference is less than 5.71 seconds, based on the finding that the average number of contact events per 80 minutes of rugby match play is 456.8 [18]. The desired outcomes of event flags are displayed in the third graph in Figure 3.6.

Figure 3.6: Sample graph generated from instrumented rugby vest data (after). This displays the data points for which errors have been corrected as well as the data has been labelled.

In addition, it was necessary to preprocess the input data into the required format prior to entering as inputs into the models. Due to the existence of two primary categories of machine learning models, each designed with a specific objective which will be further elaborated upon in the subsequent topic, the utilisation of the entire dataset as model inputs may result in a decrease in model accuracy or the occurrence of overfitting. This is because of the inclusion of both relevant and irrelevant features (columns) within each input [6]. Therefore, the manual feature selection process was performed in accordance with the specific objectives of each model. The data columns from FSRs located in the shoulder and chest regions were used for detecting the occurrence of the impact event, while the data columns from the MPU-6050 motion tracking sensor were utilised to detect fall events. The data was then divided into three portions, following a 70-15-15 split proportions for the training dataset (70%), validation dataset (15%), and test dataset (15%) accordingly.

Afterwards, the input data and labels were partitioned into several sizes, which are segments of 12 and 24 rows, to accommodate the distinct requirements of the models. In regards to segmentation, it is necessary to ensure that each segment was assigned only a single label, resulting in the aggregation of multiple labels from each row inside a segment. This final label was obtained by allocating a value flag of 1 when there was at least one impact event (i.e., at least one flag of 1), and a value flag of 0 when no event was identified (i.e., all values are zeros). Upon completion of this step, the data was ready to be used as it was effectively cleansed and prepared into an appropriate input format for the models.

3.3 Modelling with Deep Learning Algorithms

In this study, the ensemble learning techniques and the deep learning algorithms were applied together for developing the models to fulfil the research's objectives. The Convolutional Neural Network (CNN), one of the architectures for deep learning algorithms, was used as a model structure to develop two distinct types of model. The first model type was designed to recognise impact events, while the second model type was specifically designed to recognise fall events. Each type comprises three models with unique designs for the structure. And a majority voting method, which is one the ensemble learning techniques, was then used to enhance the accuracy of the predictions and decide on the final result at the end. Since a time-series dataset is a sequence of data points collected in a specific time frame, allowing for the tracking of data changes over time, both the data point at the particular timestamp and the set of data points in a short interval are absolutely essential. Therefore, the use of an ensemble of models which were trained on inputs from numerous time window sizes will provide the most accuracy with respect to this particular type of dataset.

Through the analysis of time-series data, six CNN models were constructed to identify each category of event, three models for each type. The core structure of each model exhibits similarities, starting with an input layer and followed by multiple layers of 1D Convolution (Conv1D), Dropout, and 1D Average Pooling (AveragePooling1D) layers for the purpose of feature extraction from the given data. The Conv1D layers were implemented in groups, with each layer accompanied by the Rectified Linear Activation Unit (ReLU) as its activation function. ReLU is regularly used as the activation function in hidden layers due to its effectiveness in mitigating the issue of vanishing gradients, which commonly arises during the process of learning. The Dropout and AveragePooling1D layers were added between each set. The Dropout layer is a powerful layer that helps preventing the occurrence of the overfitting issue by temporarily disabled neuron nodes in a hidden layer in a random manner while training, while the AveragePooling1D layer contributes to improved computational efficiency and enhanced extraction of detailed information by taking the average value in each kernel. A Flatten layer was then incorporated to the model architecture to transform a multidimensional array of the feature maps into a one dimensional vector. In the final stage of the architecture of the models, the multiple dense layers were implemented as hidden layers and an output layer. All of which utilise the ReLU activation function except for the output layer. The output layer was applied with the Sigmoid activation function, resulting in predicted values ranging from 0 to 1. These values can be interpreted as probabilities and mapped to binary classes (0 for events without impact and 1 for events with impact) using a threshold of 0.5. Apart from the core structure, the models were also applied the optimizer and early stopping configurations to prevent the overfitting issue [60].

Although each model has a similar underlying structure as stated earlier, it is important to address the differences that exist. The most significant difference is the input shape. This research specifically explored three different sizes of input shapes in order to accommodate for the characteristics of the obtained data. To elaborate, the instrumented rugby vest was designed to transmit data at an average rate of 24 times per second, which may differ depending on the wireless connection. As a result, the input sizes were separated into 1, 12, and 24. With the design of a row by row input shape, the models were able to learn whether what truly occurred in each row, however, by inputting a segment of 12 rows and 24 rows help the models acquired knowledge whether what occurred in 0.5s and 1s accordingly. Through the use of multiple input sizes in this analysis, the knowledge could pertaining to the data characteristics of both partial impact events and the duration of complete impact events were accumulated. The second difference is the models' architecture after flattened. Due to the different of the expected outcomes of each model type, the models for extracting impact events were split into two branches. This splitting was based on the anticipated outcomes of this type of models, which were the prediction of the occurrence of impact events and the specific body region, either the shoulder or chest, that is affected by impact. On the other hand, there is only one branch for fall detecting models. In addition to that, there are minor differences in the layer configurations of each model as well. The configurations were chosen according to the utmost precision achieved throughout the fine-tuning procedure. These configurations include both the number of filters and kernel size in Conv1D layers, as well as the number of filters and kernel size in Dense layers. Refer to the Appendix B for model structures in their full form.

3.4 Dashboard Creation

Aside from the event detection, one of the expected outcomes of this research is the dashboard. The purpose of it is to illustrate the detection results, together with the data collected by the instrumented mouthguard. As stated in the objectives, the target audience of this dashboard are rugby coaches. Its purpose is to facilitate the monitoring of players' discomfort and aid in the decision-making process about player replacements. In consideration of further incorporating this project into the MLDB, the code responsible for synchronising the results with the dashboard will be incorporated into the same Jupyter Notebook file with the production code for detecting events of the machine learning models. This approach will facilitate the migration and processing of the code on MLDB. In addition to the Python code, supplementary files such as template files are necessary for the generation of the dashboard.

Multiple python libraries, including HTML, CSS, and JavaScript, were used during the dashboard development process to generate its components. Starting with the dashboard template, HTML, CSS, and JavaScript were used to design the dashboard's core structure to display the outline of each component and designate user-interactive actions. This template was set aside and never be directly interacted with by end users, however, it was interacted with by the Python code, which was programmed to read through this template file and generate a new HTML file as an output file for the users' dashboards. During the developing phase, the illustration components were also generated. These consists of an animation of an impact that contacted the FSRs on the shoulder and chest or head as detected by the instrumented mouthguard, as well as a horizontal bar graph demonstrating the total duration of each impact. The Python library known as Bokeh was used to produce an interactive horizontal bar graph, while another well known library named the Matplotlib, was used to illustrate the recorded impact events. In addition to the dashboard creation stages, this research aims to deliver a product that is truly suitable for its intended purpose by incorporating the process of soliciting feedback from many perspectives. The interface was presented to individuals, ranging from regular researchers to specialists, for evaluation of its efficacy. The modifications were then made to the interface accordingly.

Chapter 4

Results

After applied various designs of layers and hyperparameters to fine-tuning the models, this chapter will first discuss the results generated by the final design of the CNN models and ensemble learning models. Then, the finished product of the dashboard will be discussed accordingly. The final design of each deep learning model will be referred to as the first, second, and third models for each type, in accordance with the order they are listed in the Appendix B.

Among the three models used for impact event detection, the first model performed outstanding results in terms of minimal loss and maximal accuracy across all models, except for the loss from shoulder impact event detection as shown in Table 4.1. The optimal validation loss for chest impact detection was around 0.0133, which the first model achieved, as well as the highest validation accuracies, which were 100% and 99.63% for shoulder impacts and chest impacts, respectively. The second model gained the most optimal validation loss of roughly 0.0108 when predicting shoulder impacts. Upon conducting a more thorough analysis of the learning curves for each model, it is observed that while the validation losses for the first and second models in Figure 4.1 and Figure 4.2 appear to be satisfactory, indicating the absence of overfitting, their performances did not exhibit significant improvements as they remained consistent. However, both second and third models possess small fluctuations in their loss values and, moreover, Figure 4.3 clearly demonstrates that the third model displays a notable issue of overfitting. This can seen by a noticeable spike in the latter epochs, despite the incorporation of multiple Dropout layers and Regularisation techniques.

Deep Learning Model	Validation Loss			Validation Accuracy	
	Total	Shoulder	Chest	Shoulder	Chest
First Model (Input Width: 1)	0.0427	0.0128	0.0133	100.00%	99.63%
Second Model (Input Width: 12)	0.0835	0.0108	0.0176	99.61%	99.61%
Third Model (Input Width: 24)	0.3811	0.0330	0.0794	99.24%	99.24%

Table 4.1: The validation losses and accuracies of impact event detection

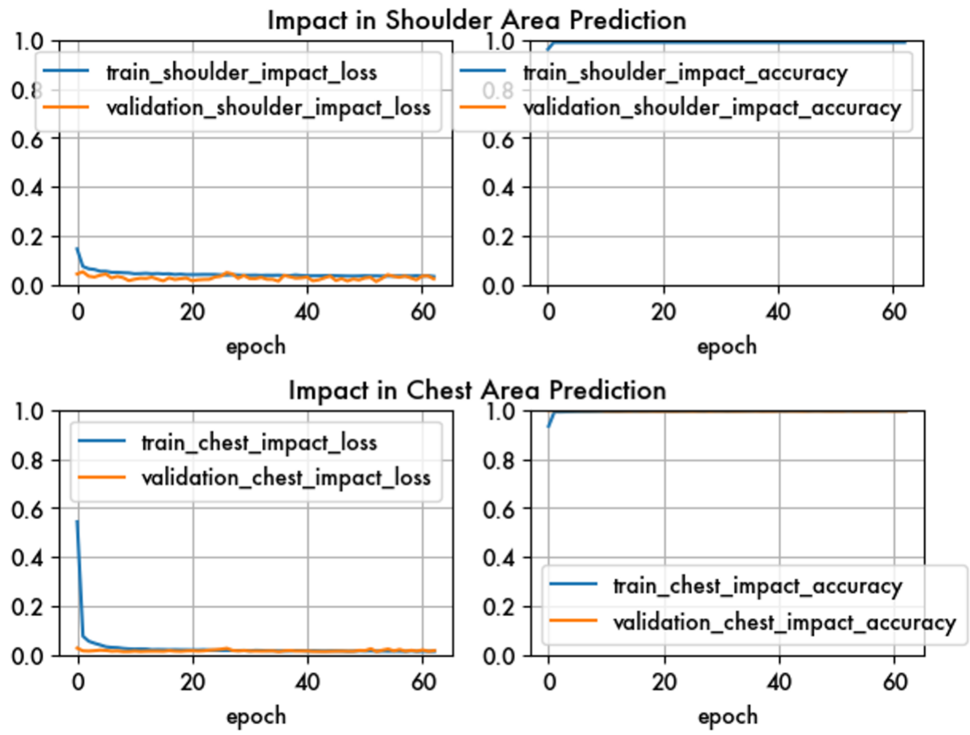

Figure 4.1: Learning curves of impact event and area of impact prediction from the first deep learning model of impact prediction type with input length equals to 1

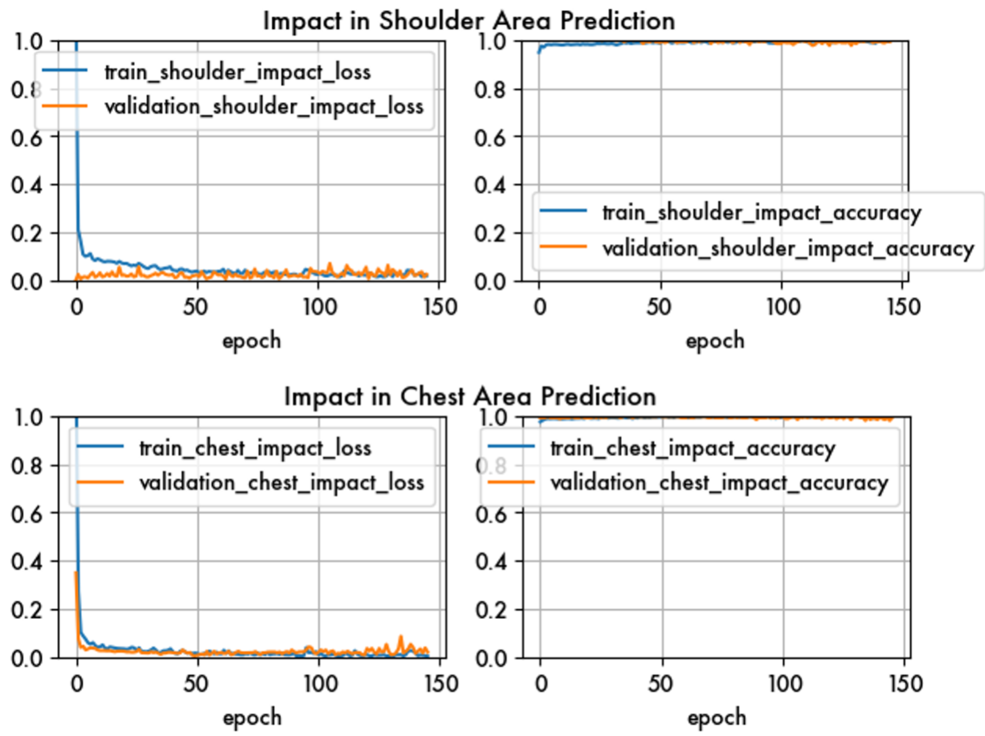

Figure 4.2: Learning curves of impact event and area of impact prediction from the second deep learning model of impact prediction type with input length equals to 12

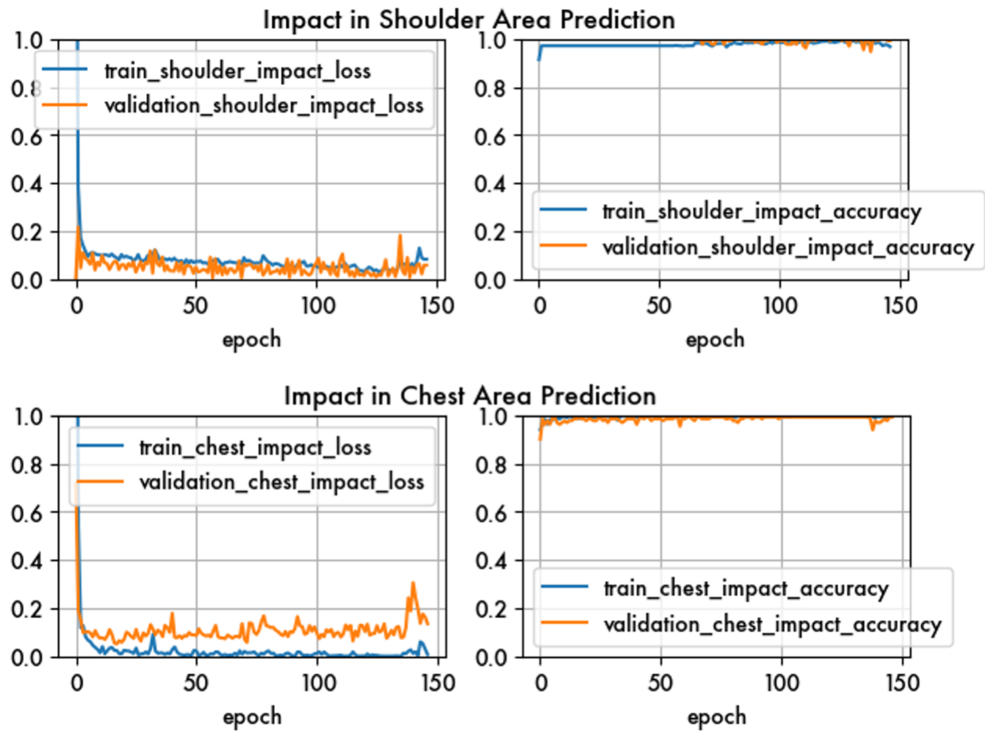

Figure 4.3: Learning curves of impact event and area of impact prediction from the third deep learning model of impact prediction type with input length equals to 24

Comparing the test phase to the validation phase, the majority of values are predictable, as test phase values were aligned with validation phase values. Only the shoulder loss that was not aligned with the validation phase as the lowest value was also achieved by the first model same as other values, while the second model was the one who achieved it in validation phase. When comparing the test phase to the validation phase, the majority of values are consistent across the two phases, indicating a strong alignment between the test phase values and the validation phase values. The misalignment of the shoulder loss during the test phase was seen to have the lowest value in the first model, but the second model achieved this lowest value during the validation phase. Nevertheless, all three models achieved accuracies over 97% in predicting shoulder and chest impact events, indicating an outstanding level of precision. All losses and accuracies were reporting in Table 4.1 and Table 4.2 accordingly.

Deep Learning Model	Test Loss			Test Accuracy	
	Total	Shoulder	Chest	Shoulder	Chest
First Model (Input Width: 1)	0.0643	0.0238	0.0239	99.63%	99.44%
Second Model (Input Width: 12)	0.2020	0.0768	0.0700	98.83%	98.44%
Third Model (Input Width: 24)	0.5149	0.1017	0.1445	96.95%	97.71%

Table 4.2: The test losses and accuracies of impact event detection

According to the data presented in Table 4.3, in terms of fall detection, the model with the highest validation accuracy was the last model of the three models for detecting falls considered. It obtained a maximal accuracy of 99.24%, while the second model achieved barely any difference at 99.22%. The first model performed slightly lower than the other two with an accuracy of 98.87%. The second model produced an ideal validation loss value of roughly 0.0338. Regarding Figure 4.4 to Figure 4.6, there is no evidence of either underfitting or overfitting in any of the models, and they all demonstrated remarkable accuracy, exceeding 98%.

Deep Learning Model	Validation Loss	Validation Accuracy
First Model (Input Width: 1)	0.0390	98.87%
Second Model (Input Width: 12)	0.0338	99.22%
Third Model (Input Width: 24)	0.0406	99.24%

Table 4.3: The validation losses and accuracies of fall event detection

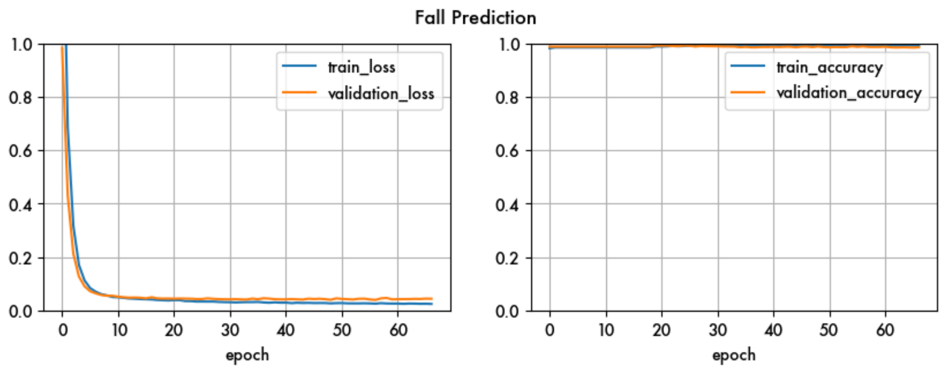

Figure 4.4: Learning curves of fall event prediction from the first deep learning model of fall prediction type with input length equals to 1

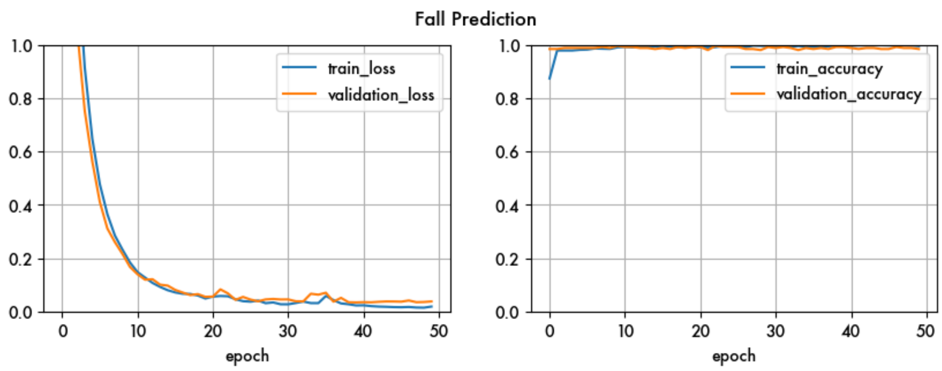

Figure 4.5: Learning curves of fall event prediction from the second deep learning model of fall prediction type with input length equals to 12

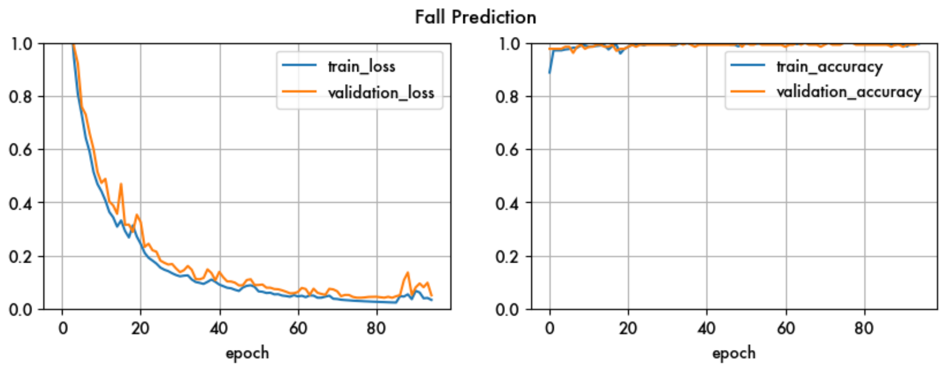

Figure 4.6: Learning curves of fall event prediction from the third deep learning model of fall prediction type with input length equals to 24

In accordance with the shoulder impact detection, an evaluation of a fall event detection during the test phase were consistent with the findings seen in the validation phase. Specifically, the second model produced the lowest test loss, while the third model gained the best test accuracy. The results for the test loss and test accuracy are shown in Table 4.4.

Deep Learning Model	Test Loss	Test Accuracy
First Model (Input Width: 1)	0.1668	96.78%
Second Model (Input Width: 12)	0.1369	96.48%
Third Model (Input Width: 24)	0.3571	96.95%

Table 4.4: The test losses and accuracies of fall event detection

After training and evaluating each of the six models, the ensemble learning method was employed by combining the results of the three models within each type using majority voting. The altered precisions are displayed in Table 4.5. The ensemble validation accuracy for predicting chest impacts was enhanced to 99.70%, while ensemble test accuracy was decreased by 0.04% from the greatest test accuracy of the chest prediction from all three models. On the other hand, both validation and test accuracies of shoulder impact detection adhere to the the greatest model-obtained accuracy. Surprisingly, both the validation and test accuracies of fall event detection were reduced to be lower than the model's lowest accuracy.

Type of Detection	Validation Accuracy	Test Accuracy
Shoulder Impact Detection	100.00%	99.63%
Chest Impact Detection	99.70%	99.40%
Fall Detection	98.77%	96.18%

Table 4.5: The accuracies of ensemble model

After the event detection process is completed, the outcome of the analysis will be presented in the form of a data visualisation tool known as a dashboard. This feature enables users to efficiently grasp the information presented. As stated, the primary objective of the dashboard was to provide real-time monitoring of players' performance throughout the rugby match, enabling the coach to make informed decisions on player replacement. The dashboard was carefully designed with a straightforward design and minimal colour palette, leveraging principles from the findings discovered in the literature review and feedback gathered from common researchers to domain expert. The overall background of the entire dashboard consisted of a light blue-grey palette, chosen to represent a sense of calmness and soothing when monitoring the dashboard amongst the intense environment of a rugby game. The use of the light blue colour was seen in the presentation of the player profile and the ranking of players within the rugby team, using an interactive interface for the purpose of selecting and monitoring the desired player. Conversely, the grey colour was employed as a frame in the section dedicated to reporting the results. This section was further divided into two primary sections, which are the Summary and Specific Details, as shown in Figure 4.7.

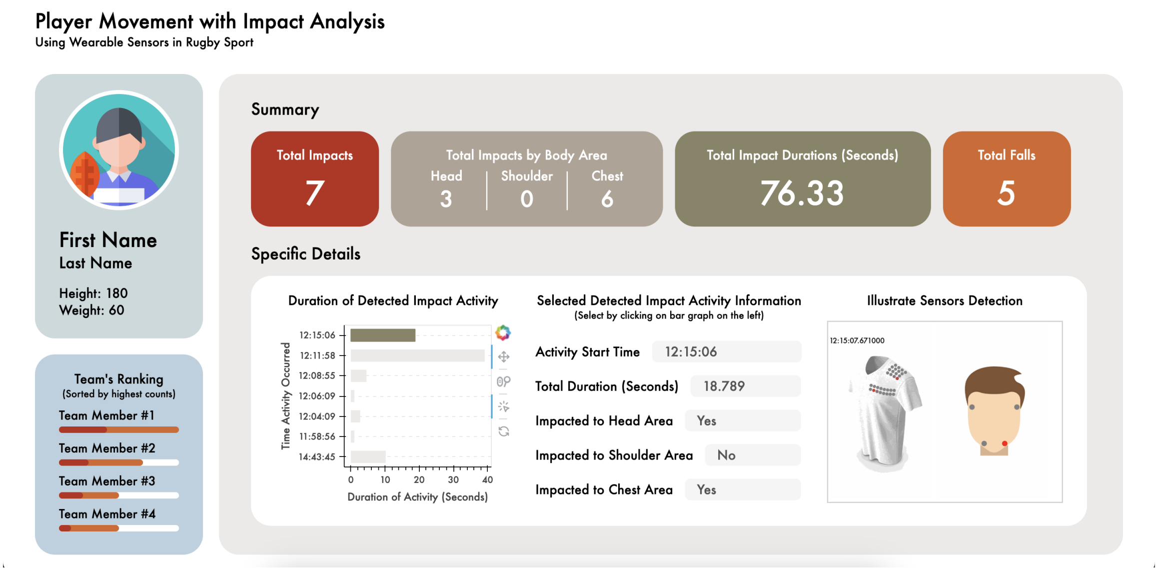

Figure 4.7: Final design of the dashboard

The summary section presents a total counts of impact and fall occurrences, together with the counts of impact affected to each body area and the total duration of impact events. The total number of impact and fall incidents was visually shown using the colours red and orange, respectively, in order to draw attention and indicate the seriousness, since they have the potential to result in injuries. Hence, the section dedicated to specific details was displayed in white colour. This choice was made to enhance readability and mitigate the potential burden of excessive information. In the initial design, a data table was presented to provide a comprehensive overview of each impact, encompassing all relevant details. However, in response to feedback and to enhance user interaction, this approach was revised. The data table was replaced with textboxes and animated GIFs, which specifically correspond to the user's interaction via the horizontal bar graph. This modification ensures that only the selected information is displayed, simplifying the presentation of data. In addition to this, several labels were also modified to enhance comprehensibility and attractiveness in response to the input. Refer to the Appendix C for a visual representation of the dashboard of previous editions.

Chapter 5

Discussion

As indicated, the purpose of this study is to evaluate the effectiveness of the prototype version of the instrumented rugby vest and improve team performance, together with player safety and well-being, by utilising wearable sensors and machine learning algorithms. To achieved this objective, rugby was selected as the main field of analysis in this study due to its characteristics includes a wide range of contact motions, which consequently results in a high incidence of injuries [18]. The data was collected by a team member performing tackles with the punch bag. In total of 14 tackles were collected. With this methodology, there was a data limitation issue occurred due to the limited number of collisions in order to avoid the potential occurrence of injuries, movement actions, and individuals undertaking an action. Further details about the issue will be elaborated more upon in the Challenges and Limitations section.

Regarding the Results chapter, the final outcomes presented in six machine learning models with CNN architecture and the dashboard were developed with refined configurations in this study. It can be seen that these models exhibited satisfactory performance in accordance to the requirements of the stated purpose in this study. Each model attained at least 95% accuracy in both the validation and testing phases. The observed outcome aligns with the expected result which supports a finding in the research paper found during the process of literature review [9]. Despite the remarkable results, some models' results displays a sign of overfitting, although the early stopping method was applied as per the theoretical guidelines to mitigate this concern [60]. There are many potential causes for this issue, such as the presence of a very complex model architecture or an insufficiently large training dataset. In this study, it was observed that the limited range of data and the small size of the training dataset appear to have a significant impact on this occurrence. This result in the model being overly fitted to the training data, hence reducing its ability to generalise effectively to unfamiliar data. Apart from that, the ensemble learning method was applied with an intention to enhance the prediction accuracy. However, the accuracies of some event detection were enhanced, while others did not exhibit any improvement. This finding points out that the base models may not have received sufficient training, despite achieving an accuracy level of over 95%. This issue has been encountered in prior research, where the effectiveness of enhancing the ensemble model's performance is most evident when it comprises accurate and diverse individual learners [61]. After thorough analysis conducted, it revealed that the validation and test datasets, after being split, contained a significantly low number of events and predominantly consist of stable stages. Consequently, this imbalance leads to high accuracy scores in each training epoch, allowing early stopping to terminate prematurely. As a consequence, the model has not been properly trained, while maintaining low loss and high accuracy. This model will therefore perform inadequately in actual prediction, however, the goals of this project were still be fulfilled, as it has been demonstrated that this prototype version of the instrumented rugby vest can be used to collect pressure data and others which further refined by utilising machine learning algorithms.

Regarding the user interface, a dashboard was developed by providing HTML, CSS, and JavaScript file templates as input, and the python code will process the final product and incorporate the data into it. The Jupyter Notebook was used to generate all code and serve as a data server for this dashboard throughout development. The objective of this code structure design was to provide seamless integration with MLDB in the future. The purpose of this dashboard is for the rugby coach to monitor the impact and fall events of each player in order to make substitution decisions, as these events may result in injuries. The previous stage's outcome data was plainly displayed in both numeric and interactive graph format. However, even though feedback was obtained from a domain expert, a professional rugby coach should be presented with the product on the pitch in order to evaluate its actual efficacy. GIFs have been made for illustrating the way in which impact is associated with sensors and their respective locations. Nevertheless, it seems to be confusing regarding the continuous looping and playback of the GIF file. It would be more advantageous in this particular context to switch to an alternative file format that supports rewinding and playback. Since it is a web-based dashboard, screen size should also be considered. The responsive web configurations were configured to prevent the component from appearing unorganised when viewed on various devices. With this final version, components were added to convey the purpose of this study explicitly. The chosen colours preserve its meaning and make it easier for the intended audience to absorb the data quickly. Regarding the feedback, the design was modified, such as the label or positioning of the components. And with this dashboard, the objective to improve team performance was accomplished as the coach was able to make prompt player substitutions, allowing the team to retain only players with fewer injuries on the pitch and reducing the potential of player injuries. Also, this could be used to display injury-related statistics for improving player's welfare once sufficient data has been collected, as well as contribute to World Rugby Federation by providing accurate data [39, 58].

5.1 Challenges and Limitations

5.1.1 Hardware Challenges

Since, this study was conducted as a pilot project of exploring the feasibility as well as effectiveness of an instrumented rugby vest, it discovered design improvement areas in the prototype version with respect to issues discussed in the methodology chapter. These issues include measurement inaccuracies and misalignment of hardware location and corresponding results. Both issues primarily arise from the manual manufacturing process, which results in a less refined final product. In addition to the error caused by the utilisation of less precision resistors, the first issue caused by the wires of some resistors contact with one another, resulting in a defect value due to a short circuit. Another factor contributing to this issue is the close alignment of each force sensing resistors in the design, which leads to limited space and accidental wire contact when the user of the wearable sensor performs any movement. But another issue exists because of the wiring of some sensors is not correctly connected to the corresponding channels of the microcontroller, leading to a misalignment between the order of the sensors and the columns of output data.

5.1.2 Data Limitation

The limited quantity and lack of diversity in the dataset bring limitations to this study. The study was constrained by many factors, including time constraints, a limited number of available equipment, and a small amount of participants with knowledge of safe tackling techniques. Thus, the analysis was limited to a total of 14 tackles, all of which were performed by a single participant who is a project team member. Furthermore, the data collection process followed a consistent pattern, capturing information from the initiation to the completion of each tackle, and shuffling the data to introduce randomness is not feasible given the characteristics of the time series data. As a result, the machine learning model may learn specific patterns from the small sample size that could lead to a drawback on the generalisability of the findings causing reduced accuracy when applied to unfamiliar data later on. In addition, the data collected by the instrumented mouthguard is automatically processed by the company's programme, resulting in the loss of information that falls below their threshold. This is beneficial for the intended audience and the researcher who intends to use only the final result, but it is difficult to utilise this data for further analysis.

Chapter 6

Conclusions

This study aims to assess the usability of a prototype instrumented rugby vest and to improve the health and wellbeing of rugby players. To achieve these purposes, a comprehensive evaluation of the existing literature was conducted and the scope of the study has been clearly defined. Hence, the objective of this project was to utilise deep learning algorithms, specifically machine learning models with CNN architecture, for the purpose of detecting impact and fall events resulting from tackles in the present study. The outcomes will be presented in a user-friendly dashboard format, allowing end users to efficiently extract and interpret information. The intended audience for this study is rugby team coaches who can potentially benefit from the findings as the study aims to provide coaches with an ability in monitoring the occurrence of impact and fall events among players in their team. Through the process of recognising such occurrences, coaches can promptly substitute players who may be experiencing the effects from high impact, since these occurrences have the potential to lead to injuries or a decrease in the team's overall performance during competitive matches.

As a result, a total of six CNN models were built, with three specifically designed for detecting impact events and determining the impacted region, while the other three focused on identifying fall events. Each model achieved a high level of accuracy. Hence, the use of ensemble learning techniques was employed to additionally enhance the accuracy of the prediction, resulting in unexpected accuracies as some were increased and some were dropped. Furthermore, a visual representation in the form of a dashboard was developed as the final product to showcase the outcomes of the models. It was carefully designed. It was carefully designed in order to best utilise the components and colours and provide a pleasant experience for users. Feedbacks on the usability were gathered from individuals with varying levels of expertise.

Throughout every phase of this study, various challenges and limitations have been encountered, leading to the realisation of research gaps and the potential topics that deserves further exploration. The things are listed in the following topics.

6.1 Future work

6.1.1 Enhancement of Hardware Implementation

Despite the fact that this instrumented rugby vest prototype has been refined over time into this version and this study has shown its usefulness for the purpose of detecting wearer movements and impacts, thereby potentially reducing the possible incidence of injuries, it still has certain areas for improvement that could be addressed in a future design to deliver as an actual selling product. The potential for improvement that have been identified include three key areas: the reduction of controller size, the enhancement of accuracy in force measurement devices, and the expansion of the detecting area. After consulting with a highly skilled technician, it was determined that the FSR is responsible for the instrumented rugby vest's high sensitivity which causes error data, particularly when the user engages in movement that causes folds. To resolve this issue, it could be changed to use the customised Velostat instead which is a polymer-based sensor [12]. However, if the FSRs are retained in the next design, the location of resistors placed to the rugby vest could be expanded for measuring the impact over a broader region. Additionally, the current prototype's controller, which sufficiently large to cause moderate discomfort and sufficiently hefty to generate centrifugal force that may result in it falling out of a pocket or causing tension on the wire, may be reduced in size by using the microcontroller technology instead. In addition, it is worth noting that earlier research have identified many kinds of tiny sensors, such as Heat Flux Sensors which utilising for monitoring core body temperature, that might be potentially used in the next design of the rugby vest. By including these sensors, the measurement scope can be expanded, emphasising the study's objective of enhancing player safety and mitigating the occurrence of player injuries [31].

6.1.2 Expansion of Data Collection

In terms of the data collection, the inclusion of more varieties of sensors has the potential to enhance the broadness of analysis and improve the precision of findings in this research. Even though two distinct kinds of wearable sensors were used in this study, it is important to keep in mind that one of these sensors remains in the prototype stage. By incorporating additional data sources, it could be explore if it is possible to enhance the variety of data or verify the correctness of the prototype in case when there is an overlap area in data measurement.

Also, this study was focused on gathering data regarding tackling, which was conducted in a laboratory setting with only one participant and a punching bag. The data collection was conducted over a length of two days in order to maximise the quantity of data, while maintaining participant safety. With this approach, the researcher observed a shift in the data pattern, which might potentially be caused by the slight differences in the laboratory position settings of each day, as well as the speed and force produced by that team member during tackling. Hence, the facilitation of data collection from various players from live rugby matches or training sessions and the exploration of additional movements, such as scrum, maul, and collisions, provide an opportunity to include new impact patterns, therefore enhancing the comprehensiveness of the research.

6.1.3 Optimisation of Analysis Precision and Outcome Standard

In this work, the ensemble learning technique was applied with an aim to improve the accuracy of the machine learning model's results and address the overfitting problem. This issue arises from the lack of diversity in the data, the presence of noisy data, and the limited size of the training dataset. However, the utilisation of the K-Fold Cross-Validation method has been seen in several research papers, in which it was applied to perform resemblance analysis [19, 40]. This finding provides an opportunity to assess its level of precision and make comparisons between them. Although they were designed to serve distinct purposes, each of them eventually contribute to minimising the overfitting issue and improving model performance. Aside from that, due to the visualisation dashboard, it is recommended to explore further study about the use of inclusive design strategies that accommodate people with colour blindness, while yet preserving the intended message conveyed by colours for people without this condition.

Total Words Count: 9974

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Appendix A

Project Details

A.1 Project Aim

The main goal of this study is to enhance the performance of the rugby team while maximising the safety of the players through the integration of wearable sensor technologies and deep learning algorithms. This study seeks to reduce the number of injuries encountered by rugby players by identifying impact and fall occurrences through the collection of data gathered by multiple wearable sensors. Regarding the wearable sensors used in this project, this study also aims to evaluate the usability and performance of the instrumented rugby vest prototype, which was created by a technician at the University of Bath and will be used alongside with an instrumented mouthguard to collect movement and impact data from rugby players. The instrumented rugby vest will be used to monitor tackle impact and movement, whereas the instrumented mouthguard will collect data on linear and rotational acceleration. The anticipated outcomes of this study are the automatic detection of impact and fall events, as well as the creation of a data visualisation that summarises and illustrates the results, including the total counts of each event, duration of impact events, the precise anatomical regions impacted, and the simulation data captured by the sensors. This research has significant implications for rugby coaches, as it provides valuable insights that can be further used to evaluate the diminished performance of rugby players resulting from impacts and falls. Therefore, coaches could use this information to make informed decisions regarding player substitution. Aside from that, the intended design of the coding process aims to provide seamless integration with an machine learning database (MLDB) in order to facilitate future advantages.

A.2 Project Objectives

The project aims to achieve the following objectives:

- Involving the use of instrumented rugby vest and instrumented mouthguard in the data collection process, ensuring anonymity and participant safety.
- Integrating the datasets from both data sources in order to analyse the insights with optimum efficacy.
- Performing data pre-processing procedures to mitigate hardware-related defects

present in the dataset and to ensure that the data is appropriately formatted for input into deep learning models.

- Developing deep learning models to assess the data obtained from players' movements, with the aim of accurately predicting the occurrences of collisions and falls at particular time, along with other valuable information.
- Training and fine-tuning the deep learning models, and assessing their performance throughout the training, validation, and testing phases.
- Presenting the insights in the form of an interactive dashboard, which is designed to be advantageous for rugby coaches who are the intended recipients of this initiative.

In order to accomplish these objectives, the expected outcomes will be presented in the project requirement section, and a comprehensive review of relevant literature and product development will be conducted.

A.3 Project Plan

A.3.1 Gantt Chart

The gantt chart in Figure A.1 illustrates the project's planning, beginning with the creation of the project proposal, continuing with the literature, technology, and data survey, and submitting ethics paperwork until the dissertation is completed. The following are the projected timelines:

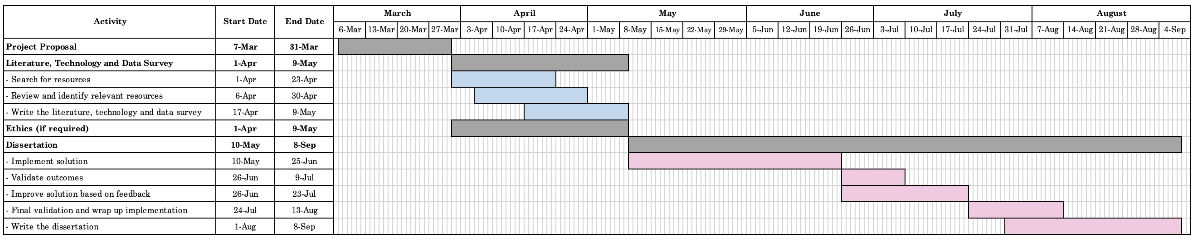

Figure A.1: Gantt chart to represent the project plan for this research

- Draft, discuss and create the project proposal from May 7th, 2023 to May 31st, 2023.
- Conduct the literature, technology and data survey from April 1st, 2023 to May 9th, 2023. This includes the following subtasks:
  - Search for previous studies from April 1st, 2023 to April 23rd, 2023.
  - Review and identify the relevant studies from April 6th, 2023 to April 30th, 2023.
  - Draft, discuss and finalise the literature, technology and data survey from April 17th, 2023 to May 9th, 2023.
- Discuss and submit the ethics paperwork from April 1st, 2023 to May 9th, 2023.

- Conduct the dissertation from May 10th, 2023 to September 8th, 2023. This includes the following subtasks:
  - Collect data and implement solution from May 10th, 2023 to June 25th, 2023.
  - Discuss and validate outcomes from June 26th, 2023 to July 9th, 2023.
  - Improve the outcomes based on the discussion feedback from June 26th, 2023 to July 23th, 2023.
  - Final discussion and finish implementation from July 24th, 2023 to August 13th, 2023.
  - Summarise and finish writing the dissertation paper from August 1st, 2023 to September 8th, 2023.

A.3.2 Tasks Completion Date

The following are all the key dates for this project:

- The project proposal is due on Friday 31st March 2023.
- The literature, technology and data survey is due on Tuesday 9th May 2023.
- The ethics document is due on Tuesday 9th May 2023.
- The dissertation is due on Friday 8th September 2023.

A.4 Project Requirements

A.4.1 General Functionalities

In order for this project to be considered accomplished, the functionalities stated below must be met.

- The datasets from two wearable sensors must be integrated into one.
- Data must be collected and kept securely and anonymously.
- For the best result, an appropriate machine learning approach must be explored and identified.
- To analyse and assess player movements, a machine learning model must be constructed with Python or related libraries, e.g., TensorFlow and Scikit-learn.
- Individual insights, such as player's movement and sensor pressure measurement while tackling, must be retrieved.
- The insights must be demonstrated in the form of visualisation with a suitable design, e.g., graphs or dashboard.

A.4.2 Expected Outcomes

The anticipated results of this research may be categorised into two components as follows:

- The development of deep learning models that can effectively predict to identify impact and fall events.
- The creation of a dashboard that is intended to be used as a tool for summarising the findings and presenting them to the target audience, which is rugby coaches in the context of this research.

A.4.3 Functionality Not to be Considered

While this study generated insights by using data analytics and machine learning approaches based on the instrumented rugby vest and the instrumented mouthguard, biological knowledge such as muscle damage and medical knowledge such as injury diagnosis will not be gathered, predicted, or provided.

Appendix B

Machine Learning Model Architectures

B.1 Impact Event Detection Models

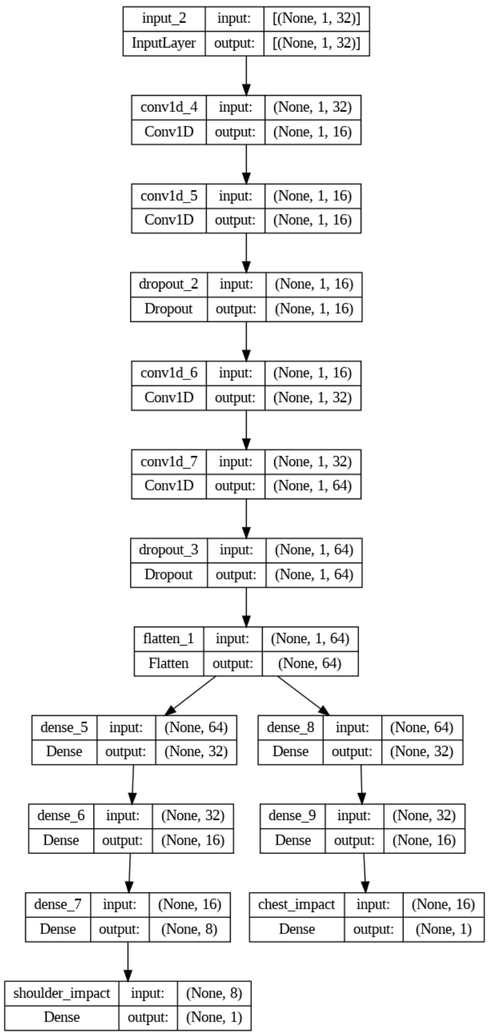

Figure B.1: Architecture of the first model used for detect impact events and impact affected body region

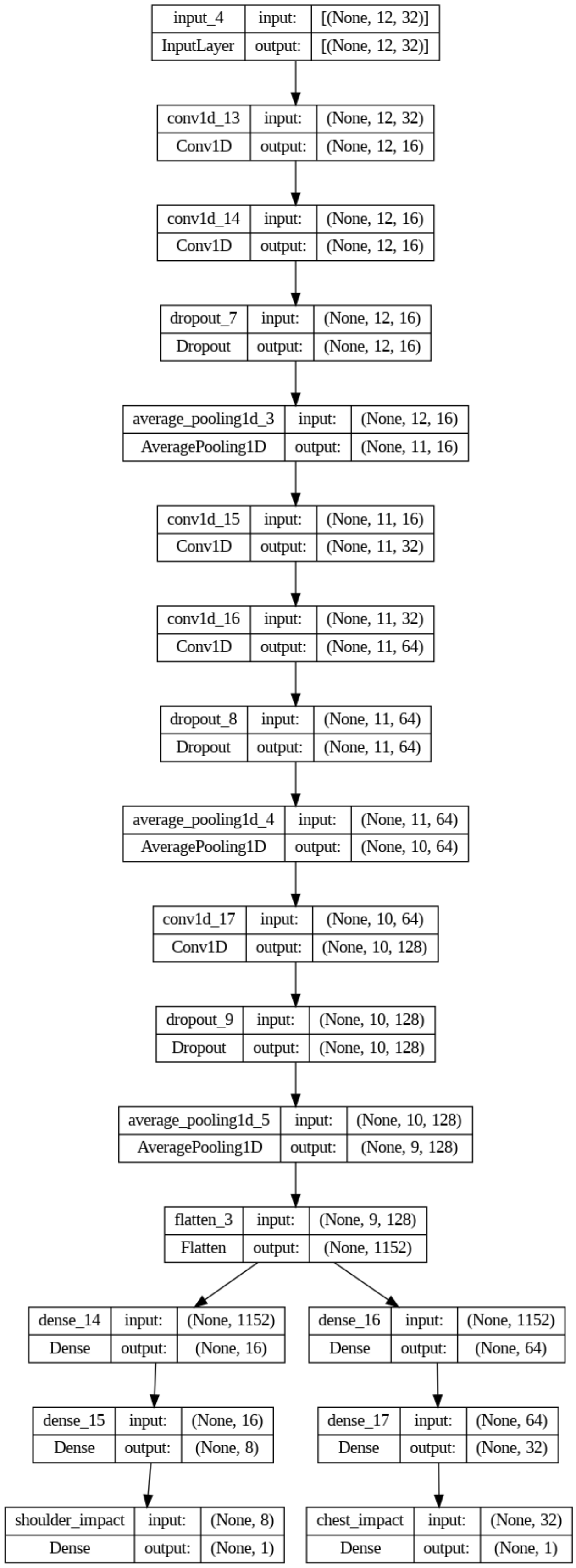

Figure B.2: Architecture of the second model used for detect impact events and impact affected body region

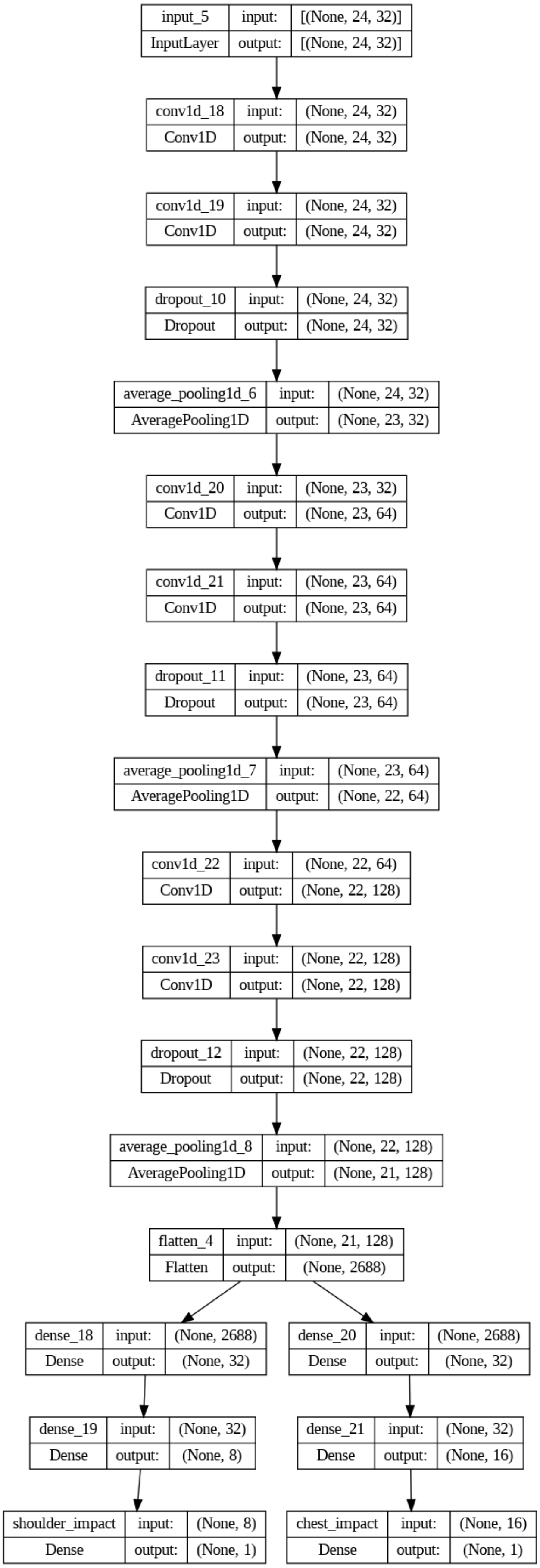

Figure B.3: Architecture of the third model used for detect fall events

B.2 Fall Detection Models

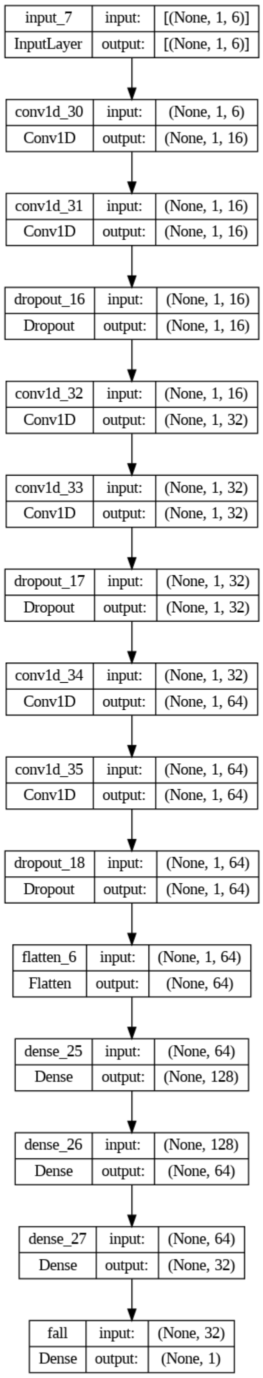

Figure B.4: Architecture of the first model used for detect falls

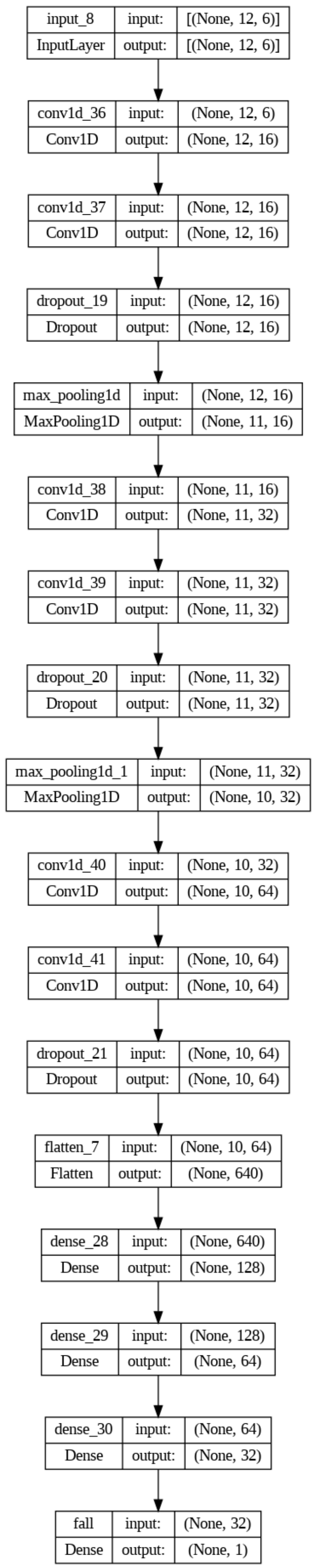

Figure B.5: Architecture of the second model used for detect falls

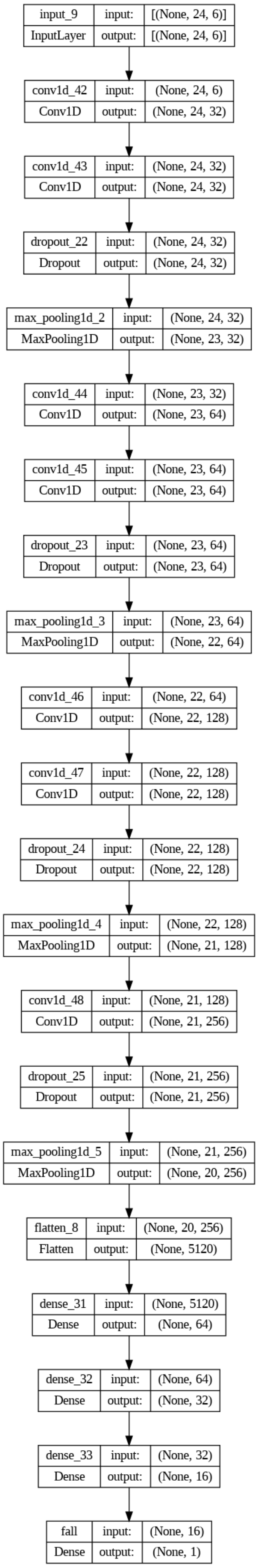

Figure B.6: Architecture of the third model used for detect falls

Appendix C

Dashboard Evolution

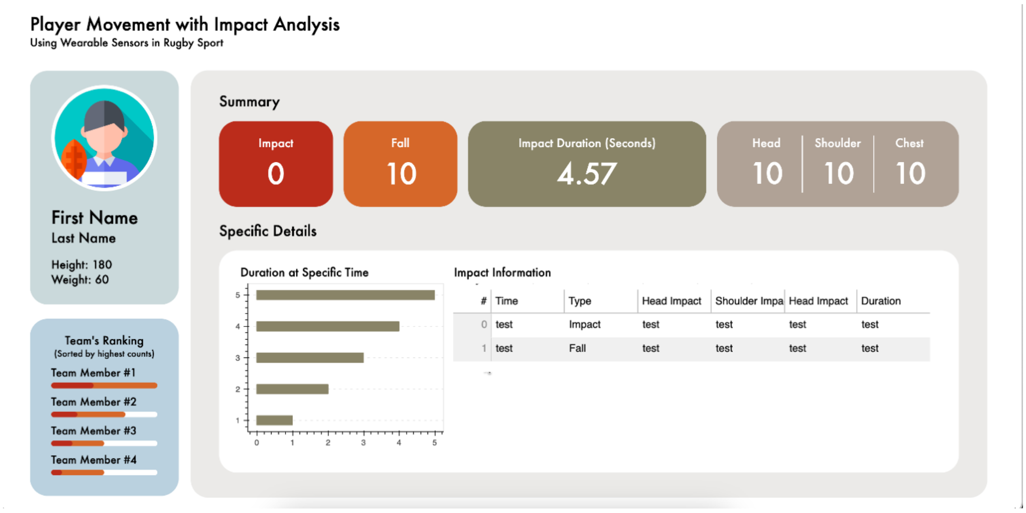

Figure C.1: The first version of dashboard before getting feedback

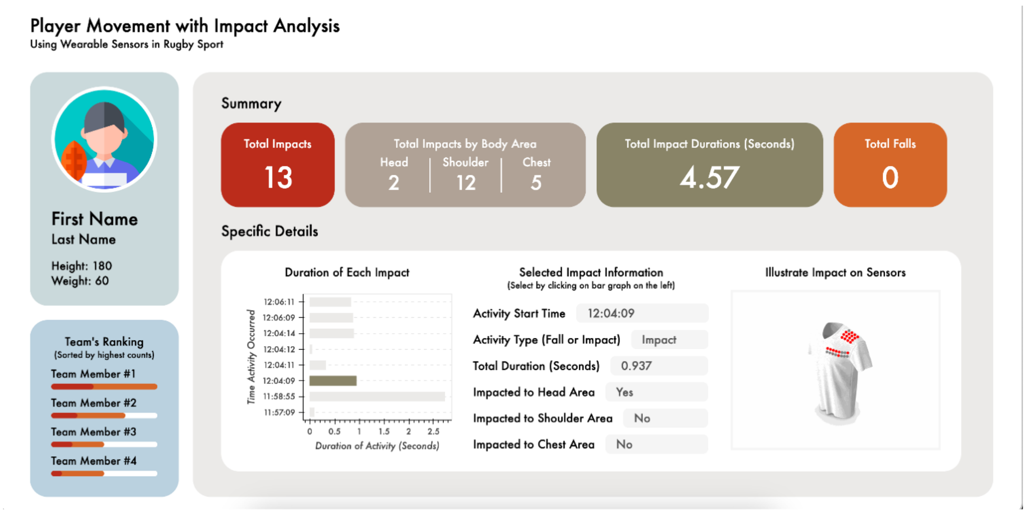

Figure C.2: The second version of dashboard after revising follows the feedback